

AI/ML Internship – Day 50 Notes & Tasks

Module 6: Building a Complete Custom YOLO Project (Helmet Detection Project)

Part 1: Recap of Day 49

Yesterday we learned:

-  Epochs
-  Loss
-  Precision
-  Recall
-  mAP
-  Training Results Analysis

Today we will combine everything learned so far and build a complete YOLO project.

Part 2: What is a Complete YOLO Project?

A complete YOLO project includes:

Problem Identification



Data Collection



Annotation



Dataset Preparation



Model Training



Model Evaluation



Testing



Deployment

Part 3: Project Example – Helmet Detection System

Problem Statement

Detect whether workers are wearing safety helmets.

Why is this Important?

Industries require workers to wear helmets.

Manual monitoring is difficult.

YOLO can automatically detect helmets using cameras.

Part 4: Project Requirements

Hardware

- Laptop/Desktop
- Webcam (Optional)

Software

- Python
- YOLOv8
- OpenCV

Dataset

Helmet Images

Part 5: Step 1 – Data Collection

Collect images containing:

Helmet

Worker Wearing Helmet

No Helmet

Worker Without Helmet

Recommended Dataset Size

Project Level Images

Beginner 100+

Intermediate 500+

Industry 2000+

Part 6: Step 2 – Annotation

Use tools:

- Labellmg
- Roboflow
- CVAT

Example

Draw a box around helmet.

```
-----  
|  HELMET  |  
-----
```

Save annotation.

Part 7: Step 3 – Dataset Structure

helmet_dataset/

└─ images

| └─ train

| └─ val

└─ labels

└─ train

└─ val

Part 8: Step 4 – Create data.yaml

train: helmet_dataset/images/train

val: helmet_dataset/images/val

nc: 1

names:

- helmet

Part 9: Understanding data.yaml

train

Training image path.

val

Validation image path.

nc

Number of classes.

nc: 1

Only Helmet class.

names

helmet

Class name.

Part 10: Step 5 – Load YOLO Model

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

Part 11: Step 6 – Train Model

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
model.train(  
    data="data.yaml",  
    epochs=20  
)
```

📌 Part 12: What Happens During Training?

Images



YOLO



Learn Helmet Features



Update Weights



Improve Detection

📌 Part 13: Training Output

Example:

Epoch 20/20

Loss = 0.85

Precision = 92%

Recall = 89%

mAP = 91%

Interpretation

Good model performance.

Suitable for testing.

📌 Part 14: Trained Model

After training:

runs/

└─ detect/

 train/

 weights/

 best.pt

📌 Part 15: Step 7 – Test the Model

```
from ultralytics import YOLO
```

```
model = YOLO(  
    "runs/detect/train/weights/best.pt"  
)
```

```
results = model(  
    "test_image.jpg"  
)
```

```
results[0].show()
```

📌 Part 16: Example Output

Input:

Worker Wearing Helmet

Output:

Helmet → 96%

with bounding box.

📌 Part 17: Real-Time Helmet Detection

```
import cv2
from ultralytics import YOLO

model = YOLO(
    "runs/detect/train/weights/best.pt"
)

cap = cv2.VideoCapture(0)

while True:

    success, frame = cap.read()

    if not success:
        break

    results = model(frame)

    annotated_frame = results[0].plot()

    cv2.imshow(
        "Helmet Detection",
        annotated_frame
    )

    if cv2.waitKey(1) == 27:
        break

cap.release()
cv2.destroyAllWindows()
```

Part 18: Real-World Workflow

CCTV Camera



Capture Image



YOLO Model



Helmet Detection



Alert System

Part 19: Other Custom YOLO Projects

Safety Vest Detection

Detect workers wearing safety jackets.

ID Card Detection

Detect employee ID cards.

Number Plate Detection

Detect vehicle number plates.

Fruit Detection

Detect fruits in supermarkets.

Defect Detection

Detect damaged products.

Part 20: Industry Deployment Concept

Camera



YOLO Model



Detection



Database



Dashboard



Alert Notification

Part 21: Theory Questions

What is a Custom YOLO Project?

A project where YOLO is trained to detect user-defined objects.

What is data.yaml?

Configuration file containing dataset information.

What is best.pt?

Best-performing trained model.

Why is annotation important?

Because YOLO learns from annotated objects.

What is the purpose of testing?

To verify model performance.

Part 22: Interview Questions

How do you build a custom YOLO project?

1. Collect Data
 2. Annotate Images
 3. Create Dataset
 4. Configure data.yaml
 5. Train YOLO
 6. Evaluate Results
 7. Test Model
 8. Deploy
-

What are the main components of a YOLO project?

- Dataset
 - Annotation
 - Training
 - Evaluation
 - Deployment
-

Why do we use custom training?

To detect objects not present in pretrained datasets.

Tasks for Day 50

Theory

1. Explain the complete YOLO workflow.
 2. Why is annotation important?
 3. What is best.pt?
 4. What is deployment?
 5. Explain the role of data.yaml.
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Practical Task 1

Create a Helmet Detection project folder.

Practical Task 2

Design dataset structure.

Practical Task 3

Write a sample data.yaml.

Practical Task 4

Write YOLO training code.

Practical Task 5

Draw the complete project workflow diagram.